

1. BEZOUT'S THEOREM

The purpose of this is to motivate sheaf cohomology by applying it to give a simple proof of Bezout's theorem. We don't develop any sheaf cohomology though - refer to Hartshorne or elsewhere.

We work over an algebraically closed field. Bezout's theorem states the following: let D and D' be smooth projective planar curves of degree d and d' . Then they intersect (counted with the 'right' notion of multiplicity) dd' times. The first question is what the 'right' notion of intersection multiplicity is. At least to me, whatever this definition is, it should agree with the algebraic multiplicity appearing in the factorisation of a polynomial from the fundamental theorem of algebra. That is, if we look at the polynomial

$$y = (x - a_1)^{e_1} \cdots (x - a_r)^{e_r},$$

then this should intersect the x -axis $\sum e_i$ times (with multiplicity e_i at each point a_i). Reframing this geometrically, if we let

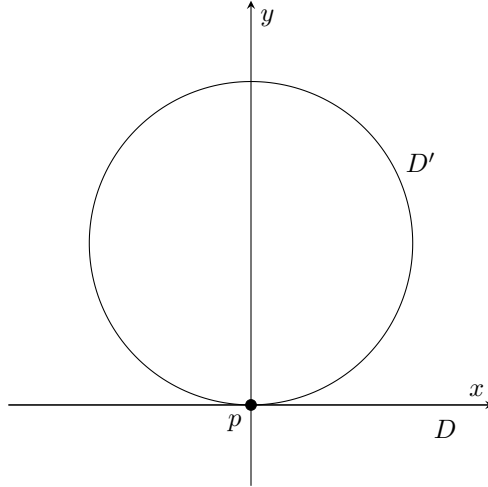
$$f = y, \quad g = y - (x - a_1)^{e_1} \cdots (x - a_r)^{e_r}$$

and

$$D = V(f), \quad D' = V(g)$$

then D and D' should intersect $\sum e_i$ times. The problem is, this doesn't generalise too well when we consider polynomials f and g such that, even after a change of coordinates, one cannot be written as polynomial function in x . For example, take

$$D = V(y), \quad D' = V((y - 1)^2 + x^2 - 1).$$



The intuitive intersection multiplicity at p should be 2, but there isn't an immediate factorisation we can use. So instead we use the fact that these are smooth curves; at every point, there is a local parameter t_p . Recall what this means: for a smooth curve X and its structure sheaf/sheaf of regular functions $\mathcal{O}_X$, the stalk $\mathcal{O}_{X,p}$ is a regular local ring of dimension 1. This means it is a DVR: its maximal ideal $\mathfrak{m}_p$ is generated by some element t_p with the property that any $h \in K(X) = \text{Frac}(\mathcal{O}_{X,p})$

can be expressed as $h = ut_p^r$, with $u \in \mathcal{O}_{X,p}^\times$. So when $D = V(f)$ and $D' = V(g)$, we try the following definition:

$$\text{mult}_p(D, D') = \text{ord}_p(f|_{D'}),$$

where $f|_{D'}$ denotes the residue class in the stalk. For the moment, we might just hope that this is well-defined. Let's see how this might look. Return to the example above:

$$D = V(y), \quad D' = V((y-1)^2 + x^2 - 1).$$

Let $p = (0, 0)$. Then

$$\mathcal{O}_{D',p} = \frac{k[x, y]_{(x,y)}}{((y-1)^2 + x^2 - 1)},$$

with maximal ideal $\mathfrak{m}_{D',p} = (x, y)$. We claim this is generated by x . Indeed, observe that if

$$(y-1)^2 + x^2 - 1 = 0,$$

then

$$y = x^2(2-y)^{-1}.$$

Since $2-y$ does not vanish at the origin, it is a unit in $\mathcal{O}_{D',p}$. So x is a local parameter, and clearly $\text{ord}_p(y|_{D'}) = 2$. We can also check if it is well-defined in this case by looking at D . We have

$$\mathcal{O}_{D,p} \simeq \frac{k[x, y]_{(x,y)}}{(y)k[x, y]_{(x,y)}} \simeq k[x]_{(x)}.$$

We can use the local parameter x , where

$$((y-1)^2 + x^2 - 1)|_D = x^2.$$

Then $\text{ord}_p(x^2) = 2$. Importantly, it is easy to check that this definition agrees with our earlier one where

$$D = V(y), \quad D' = V(y - (x - a_1)^{e_1} \cdots (x - a_r)^{e_r}).$$

Lastly, we should check this definition is well-defined. Again, let $D = V(f)$ and $G = V(g)$. Then

$$\frac{\mathcal{O}_{\mathbb{P}^2,p}^2}{(f, g)} \simeq \frac{\mathcal{O}_{\mathbb{P}^2,p}/(f)}{(g)} = \frac{\mathcal{O}_{D,p}}{(g)}.$$

Then $g = ut_p^r$, so

$$\frac{\mathcal{O}_{D,p}}{(g)} \simeq \frac{\mathcal{O}_{D,p}}{(t_p^r)},$$

which gives us the chain

$$0 \subset \frac{(t^{r-1})}{(t^r)} \subset \cdots \subset \frac{(t)}{(t^r)} \subset \frac{\mathcal{O}_{D,p}}{(t^r)}.$$

Note that

$$\frac{(t^i)/(t^r)}{(t^{i+1})/(t^r)} \simeq \frac{(t^i)}{(t^{i+1})} \simeq \kappa(p) \simeq k,$$

where the last isomorphism is from the assumption of algebraic closure. So the chain is a composition series of length r , which implies

$$\text{mult}_p(D, D') = \text{length} \frac{\mathcal{O}_{\mathbb{P}^2,p}^2}{(f, g)}.$$

Isomorphic modules have the same length, so we would get the same result from considering instead

$$\frac{\mathcal{O}_{D',p}}{(f)},$$

hence what we have been working with is well-defined.

Definition 1. The intersection number of D and D' is

$$\sum_{p \in D \cap D'} \text{mult}_p(D, D').$$

2. APPLICATION OF SHEAF COHOMOLOGY

For the rest of this section, let $D = V(F)$ and $D' = V(G)$ be smooth projective planar curves of degrees d and d' respectively. Let X be a variety over k and $\mathcal{F}$ a locally free sheaf. Then we define

$$h^i(X, \mathcal{F}) = \dim_k H^i(X, \mathcal{F}).$$

Let $Z = D \cap D'$. We claim

$$h^0(Z, \mathcal{O}_Z) = \sum_{p \in D \cap D'} \text{mult}_p(D, D').$$

To see this, note Z is finite and has dimension 0, so

$$H^0(Z, \mathcal{O}_Z) \simeq \prod_{p \in Z} \mathcal{O}_{Z,p}.$$

Hence

$$h^0(Z, \mathcal{O}_Z) = \sum \dim_k \mathcal{O}_{Z,p}.$$

Since k is algebraically closed,

$$\dim_k \mathcal{O}_{Z,p} = \text{length } \mathcal{O}_{Z,p},$$

which gives us what we are looking for.

Definition-Theorem 2. For a variety X , define

$$\chi(X, \mathcal{F}) = \sum_{i=0}^{\dim X} (-1)^i h^i(X, \mathcal{F}).$$

For $\mathcal{F} = \mathcal{O}_X(m)$, this is a polynomial in m . We define the Hilbert polynomial $P_X(m) = \chi(X, \mathcal{O}_X(m))$.

Theorem 3. For a short exact sequence on X ,

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

we have that

$$\chi(X, \mathcal{F}) = \chi(X, \mathcal{F}') + \chi(X, \mathcal{F}'').$$

Theorem 4. If $X = \mathbb{P}^n$, then

$$P_X(m) = \binom{n+m}{n}.$$

Proof. Serre vanishing implies $h^i(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(m)) = 0$ for $m \gg 0$. So for $m \gg 0$, we get

$$\chi(\mathcal{O}_{\mathbb{P}^2}(m)) = h^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(m)) = \binom{n+m}{n},$$

by a stars and bars argument. By [2], we have the polynomial equality

$$P_{\mathbb{P}^2}(m) = \chi(\mathcal{O}_{\mathbb{P}^2}(m))$$

for the infinitely many $m \gg 0$, so we get everywhere that

$$P_{\mathbb{P}^2}(m) = \binom{n+m}{n}.$$

□

Theorem 5. *For the curve D , $P_D(m)$ is linear, with leading coefficient d .*

Proof. We have the exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^2} \longrightarrow \mathcal{O}_D \longrightarrow 0$$

where the second arrow is multiplication by F and the third is restriction to D . Twisting by m is exact, so we also have the exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(m-d) \longrightarrow \mathcal{O}_{\mathbb{P}^2}(m) \longrightarrow \mathcal{O}_D(m) \longrightarrow 0$$

So

$$\chi(\mathcal{O}_D(m)) = \chi(\mathcal{O}_{\mathbb{P}^2}(m)) - \chi(\mathcal{O}_{\mathbb{P}^2}(m-d)).$$

As polynomials (using [4]):

$$\begin{aligned} P_D(m) &= P_{\mathbb{P}^2}(m) - P_{\mathbb{P}^2}(m-d) \\ &= \binom{m+2}{2} - \binom{m-d+2}{2} \\ &= dm + 1 - \frac{(d-1)(d-2)}{2}. \end{aligned}$$

□

Theorem 6. *Bezout's theorem: we have that*

$$h^0(Z, \mathcal{O}_Z) = dd'.$$

Proof. We have the short exact sequence

$$0 \longrightarrow \mathcal{O}_D(-d') \xrightarrow{\times G} \mathcal{O}_D \longrightarrow \mathcal{O}_Z \longrightarrow 0$$

where the second arrow is multiplication of a section by G and the third arrow is restriction to Z . Twisting by m is exact, so we also have the exact sequence

$$0 \longrightarrow \mathcal{O}_D(m-d') \xrightarrow{\times G} \mathcal{O}_D(m) \longrightarrow \mathcal{O}_Z(m) \longrightarrow 0$$

So (using the results from [5])

$$\begin{aligned} P_Z(m) &= P_D(m) - P_D(m-d') \\ &= dm + 1 - \frac{(d-1)(d-2)}{2} - (d(m-d') + 1 - \frac{(d-1)(d-2)}{2}) \\ &= dd'. \end{aligned}$$

So $P_Z(m) = dd' = h^0(Z, \mathcal{O}_Z(m))$.

□